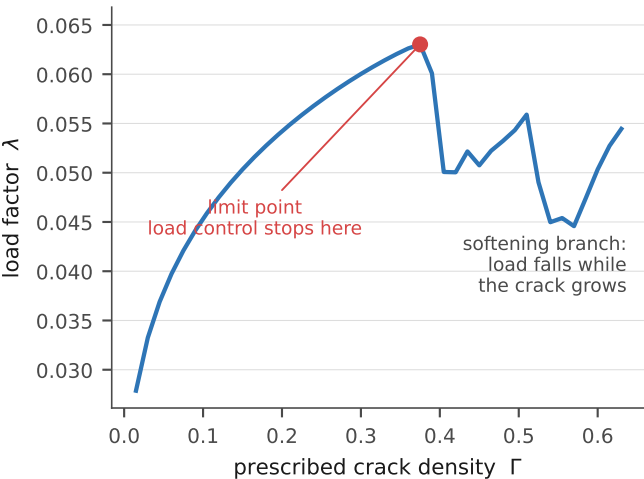


(a) crack-length control



(b) effective permeability, five loading paths

